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Winter Term 2025/2026

Foundations of Audio Signal Processing
Exercise sheet 8

To be uploaded in eCampus till: 19-12-2025 22:00 (strict deadline)

Exercise 8.1

[5 points]

The sinc-function serves as synthesis function for the sampling theorem. It is defined by

$$\operatorname{sinc}(t) := \begin{cases} 1, & \text{if } t = 0, \\ \frac{\sin(\pi t)}{\pi t}, & \text{if } t \neq 0. \end{cases}$$

Prove that the integer translates $t \mapsto \operatorname{sinc}(t-k), k \in \mathbb{Z}$, are pairwise orthogonal with respect to the inner product, i.e., $\langle \operatorname{sinc}(\cdot - k) | \operatorname{sinc}(\cdot - \ell) \rangle = \delta_{k\ell}$ for $k, \ell \in \mathbb{Z}$. Use the property that the Fourier transform defines an isometry on $L^2(\mathbb{R})$, i.e., for $f, g \in L^2(\mathbb{R})$: $\langle f | g \rangle = \langle \hat{f} | \hat{g} \rangle$ (Plancherel's theorem).

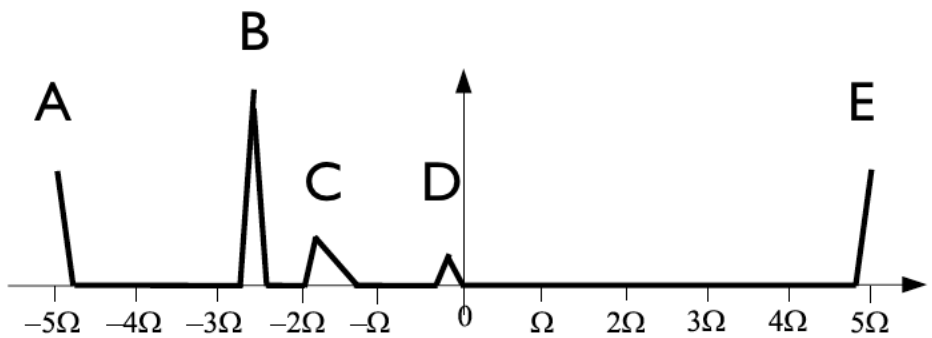
Exercise 8.2

[$4 + 4 = 8$ points]

The sampling theorem states that an Ω-bandlimited signal $f \in L^2(\mathbb{R})$ can be reconstructed from its sample values if a sample rate $\frac{1}{T} = 2\Omega$ is used. Let the continuous signal $f \in L^2(\mathbb{R}), f : \mathbb{R} \to \mathbb{C}$, be 5Ω-bandlimited. We consider the T-sampled version of f for $T := \frac{1}{2\Omega}$.

(a) Specify the Fourier transform $\hat{g}$ of the signal $g \in L^2(\mathbb{R})$ which is represented by the T-sampled version of f. Explain how you get to this result. Proceed as in the example from the lecture.

(b) Sketch $|\hat{g}|$ for the case that $|\hat{f}|$ has the following appearance:



Please add labels A-E to your sketch to indicate the resulting positions of the five non-zero spectral regions of $|\hat{f}|$.

Exercise 8.3

[$2 + 3 = 5$ points]

When you dial a number on a push-button telephone, each number is associated to a superposition of two sine waves of different frequencies. The following table indicates which frequencies are associated to which number:

	1209 Hz	**1336 Hz**	**1477 Hz**
697 Hz	1	2	3
770 Hz	4	5	6
852 Hz	7	8	9
941 Hz	$*$	0	#

This means that when dialing number 1 on the telephone, a sine wave of 1029 Hz and one of 697 Hz are generated and added.

(a) You are given a .wav file in which you can hear a person dialing a 6 digits number. Explain how you could find out which numbers have been dialed and in which order.

(b) Implement a Matlab / Python function that, given the .wav file ”dialtones.wav” (which you can find together with the exercise sheet on eCampus), helps you finding out which 6 numbers have been dialed. Note that in the given .wav file, each tone lasts 1 second. Submit the .m file and also state clearly in your solutions the numbers you found out in the correct order.
(Hint: The function does not need to output the numbers, but you should be able to ”read-off” the results from suitable plots.).